

Research Notes on Adversarial machine learning

Adversarial machine learning

>> Section 1

$S(x') > 0 \iff \begin{cases} \arg \max_c F(x') \neq C(x), & \text{(Untargeted)} \\ \arg \max_c F(x') = c^*, & \text{(Targeted)} \end{cases}$ $\{\displaystyle S(x^{\prime })>0\text{iff } \begin{cases} \arg \max _{c} F(x^{\prime })\neq C(x), & \text{\text{(Untargeted)}} \\ \arg \max _{c} F(x^{\prime })=c^{*}, & \text{\text{(Targeted)}} \end{cases}\}$

>> Section 2

Labeling and ground truth challenges Malware labels are often unstable because different antivirus engines may classify the same sample in conflicting ways. Ceschin et al. note that families may be renamed or reorganized over time, causing further discrepancies in ground truth and reducing the reliability of benchmarks.

>> Section 3

$\min_{x \in [0, 1]^d} L(f(x), y), \text{ s.t. } \|x - x^*\|_p \leq \epsilon$ $\{\displaystyle \min _{\hat {x} \in [0,1]^d} L(f(\hat {x}),y),\text{ s.t. } ||\hat {x}-x||_p\leq \epsilon \}$

>> Section 4

When solved using gradient descent, this equation is able to produce stronger adversarial examples when compared to fast gradient sign method that is also able to bypass defensive distillation, a defense that was once proposed to be effective against adversarial examples.

>> Section 5

Simple black-box adversarial attacks Simple black-box adversarial attacks is a query-efficient way to attack black-box image classifiers. Take a random orthonormal basis $v_1, v_2, \dots, v_d$ $\{\displaystyle v_{1},v_{2},\ldots ,v_{d}\}$ in $\mathbb{R}^d$ $\{\displaystyle \mathbb{R} ^d\}$. The authors suggested the discrete cosine transform of the standard basis (the pixels). For a correctly classified image x $\{\displaystyle x\}$, try $x + \epsilon v_1, x - \epsilon v_1$ $\{\displaystyle x+\epsilon v_{1},x-\epsilon v_{1}\}$, and compare the amount of error in the classifier upon $x + \epsilon v_1, x - \epsilon v_1$ $\{\displaystyle x+\epsilon v_{1},x-\epsilon v_{1}\}$. Pick the one that causes the largest amount of error.

>> Section 6

Targeted: $\min_{x'} d(x', x)$ subject to $C(x') = c^*$ $\{\displaystyle \textbf{\text{Targeted:}} \min _{x^{\prime }} d(x^{\prime },x)\text{ subject to } C(x^{\prime })=c^{*}\}$

>> Section 7

Fast gradient sign method One of the first proposed attacks for generating adversarial examples was proposed by Google researchers Ian J. Goodfellow, Jonathon Shlens, and Christian Szegedy. The attack was called fast gradient sign method (FGSM), and it consists of adding a

linear amount of in-perceivable noise to the image and causing a model to incorrectly classify it. This noise is calculated by multiplying the sign of the gradient with respect to the image we want to perturb by a small constant epsilon. As epsilon increases, the model is more likely to be fooled, but the perturbations become easier to identify as well. Shown below is the equation to generate an adversarial example where x $\{\textstyle x\}$ is the original image, ϵ $\{\textstyle \epsilon\}$ is a very small number, Δ_x $\{\textstyle \Delta _x\}$ is the gradient function, J $\{\textstyle J\}$ is the loss function, θ $\{\textstyle \theta\}$ is the model weights, and y $\{\textstyle y\}$ is the true label.

>> Section 8

Adversarial machine learning is the study of the attacks on machine learning algorithms, and of the defenses against such attacks. Machine learning techniques are mostly designed to work on specific problem sets, under the assumption that the training and test data are generated from the same statistical distribution (IID). However, this assumption is often dangerously violated in practical high-stake applications, where users may intentionally supply fabricated data that violates the statistical assumption. Most common attacks in adversarial machine learning include evasion attacks, data poisoning attacks, Byzantine attacks and model extraction.

>> Section 9

With this boundary function, the attack then follows an iterative algorithm to find adversarial examples x' $\{\textstyle x^{\prime }\}$ for a given image x $\{\textstyle x\}$ that satisfies the attack objectives.

>> Section 10

Data collection issues Security datasets vary across formats, including binaries, network traces, and log files. Studies have reported that the process of converting these sources into features can introduce bias or inconsistencies. In addition, time-based leakage can occur when related malware samples are not properly separated across training and testing splits, which may lead to overly optimistic results.

>> Section 11

Challenges in applying machine learning to security Researchers have observed that the constraints under which machine-learning techniques function in the security domain are different from those of common benchmark domains. Security data may change over time, include mislabeled samples, or reflect adversarial behavior, which complicates evaluation and reproducibility.

>> Section 12

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In other words, finding some perturbed adversarial example $\hat{x}$ such that the classifier incorrectly classifies it to some other class under the constraint that $\hat{x}$ and x are similar. The paper then defines loss L as $L(f(\hat{x}), y) = f(y(\hat{x})) - \max_{k \neq y} f(k(\hat{x}))$ and proposes the solution to finding adversarial example $\hat{x}$ as solving the below constrained optimization problem:

>> Section 13

Data poisoning Poisoning consists of contaminating the training dataset with data designed to increase errors in the output. Given that learning algorithms are shaped by their training datasets, poisoning can effectively reprogram algorithms with potentially malicious intent. Concerns have been raised especially for user-generated training data, e.g. for content recommendation or natural language models. The ubiquity of fake accounts offers many opportunities for poisoning. Facebook reportedly removes around 7 billion fake accounts per year. Poisoning has been reported as the leading concern for industrial applications. On social medias, disinformation campaigns attempt to bias recommendation and moderation algorithms, to push certain content over others. A particular case of data poisoning is the backdoor attack, which aims to teach a specific behavior for inputs with a given trigger, e.g. a small defect on images, sounds, videos or texts.

>> Section 14

HopSkipJump attack This black box attack was also proposed as a query efficient attack, but one that relies solely on access to any input's predicted output class. In other words, the HopSkipJump attack does not require the ability to calculate gradients or access to score values like the Square Attack, and will require just the model's class prediction output (for any given input). The proposed attack is split into two different settings, targeted and untargeted, but both are built from the general idea of adding minimal perturbations that leads to a different model output. In the targeted setting, the goal is to cause the model to misclassify the perturbed image to a specific target label (that is not the original label). In the untargeted setting, the goal is to cause the model to misclassify the perturbed image to any label that is not the original label. The attack objectives for both are as follows where x is the original image, x' is the adversarial image, d is a distance function between images, c^* is the target label, and C is the model's classification class label function: